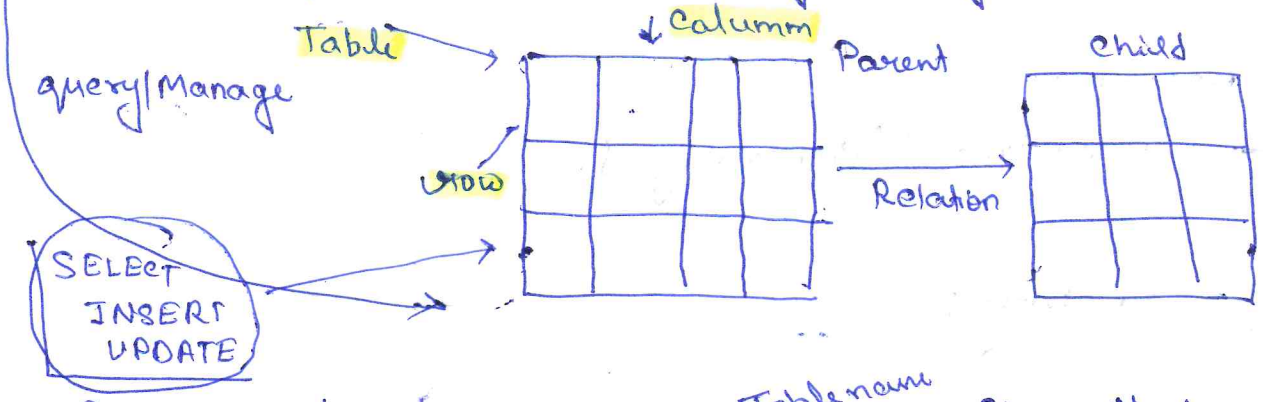
SQL Vs NoSQL

SQL & NoSQL	
1. STRUCTURE	2. NATURE
3. SCALABILITY	4. PROPERTY

SQL :- (Structured Query Language) : It is use to query Relation Databe Management System (RDBMS)

Concept&Coding



STRUCTURE

① Table, row, Column

② Predetermind Schema before you start using it.

we plan the schema before the Insert update operation

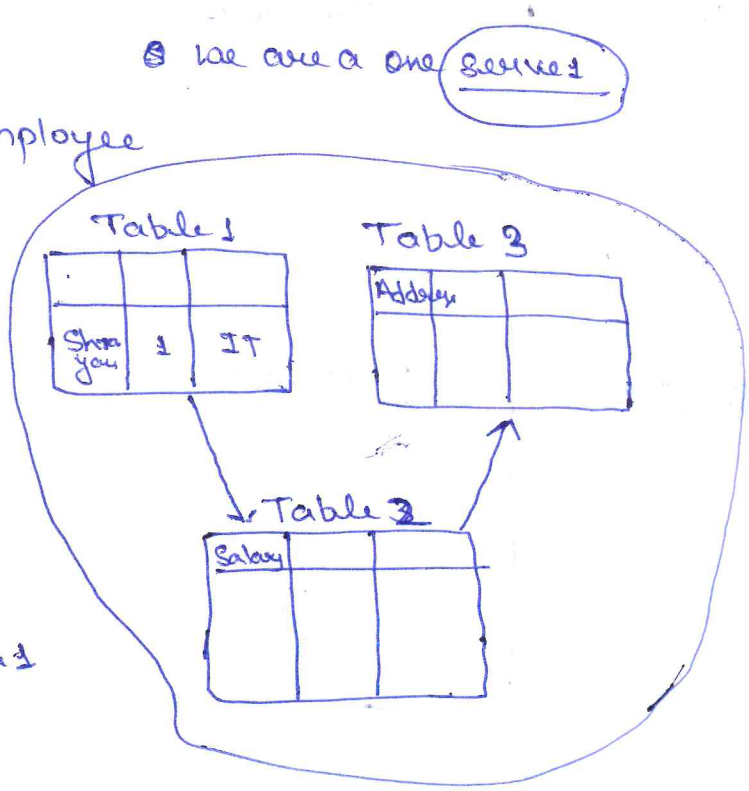
③

Tablenam

String	Number(20)
name	Roll

~~Nature of SQL :-~~

Nature of SQL :→



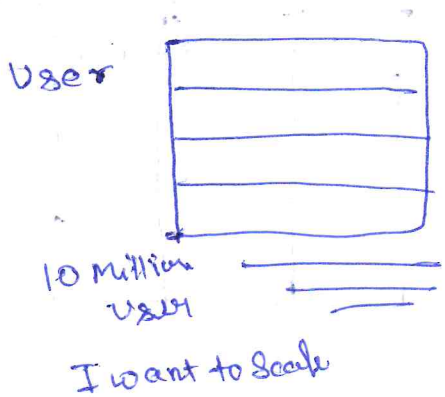
- In a particular server if for particular user or you can say that particular employee. The whole data a present into one server.

Concept && Coding

Nature :-

- Concontrated / Centralized.

3: Scalability :→



① Verticle
↑ RAM
↑ Storage Capacity

② Horizantal
Sharding
based on rows and column
Not well Supported in SQL

* Verticle much fit better to SQL.

4. Properties

ACID
- Atomocity
- Consistency
- Isolation
- Durability

make sure the Data Integerity

Consistency

NoSQL

[Non relation,
NoSQL,
Not only SQL]

1:- STRUCTURE :-

- NoSQL works on Unstructure data

Concept && Coding

1. Key-value DB
2. Document DB
3. Colum-wise DB
4. Graph DB

1. What is Key-value DB
   → Dynamo DB

Query → Search on Key

Key	Value
1	String | Integer | JSON

Opaque do the

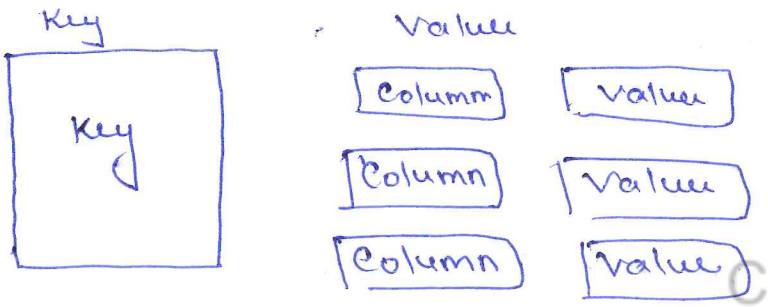
(2) Document DB

Key	Value
1	{ JSON } or { XML }

Search base on key

* Search operation base on key and we can also Query on DB. value.
  Ex: MongoDB

(3) Column-wise DB :→

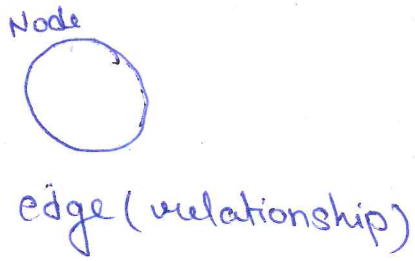


Concept && Coding

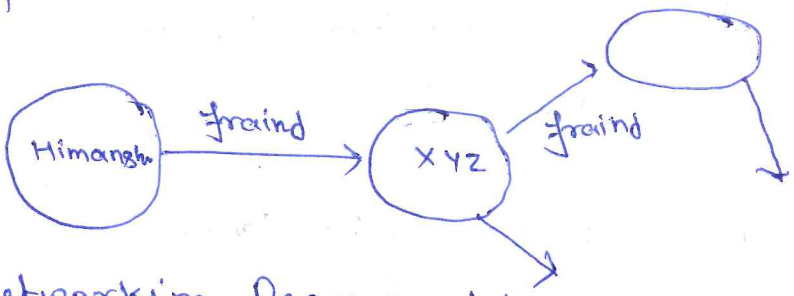
Example:

Key	Column	Value
1000 →	firstname	: Himansh
	department	: IT
	Interest	: BasketBall
1002 →	first-name	: Raj
	deptmnt	: Sales.

4 :- **GraphDB** :→ GraphDB can store the data into Node and Edge



Example

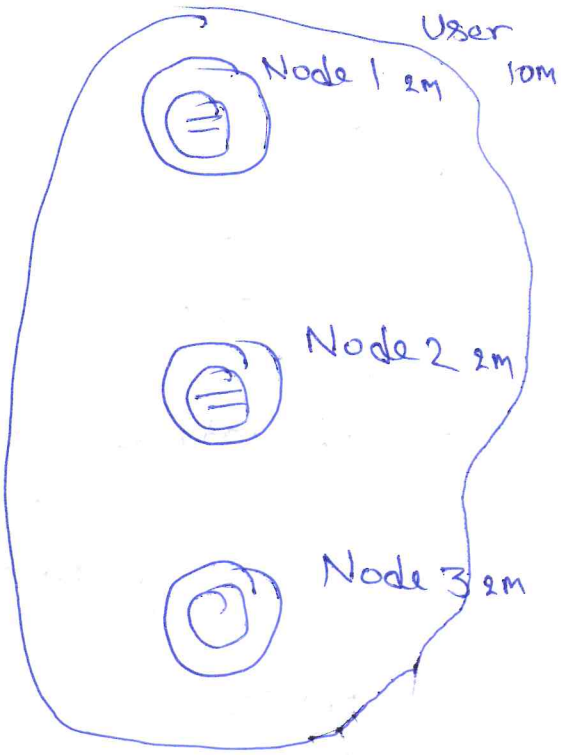


Social Networking, Recommendation Engin use this type of DB.

- It's a very very fast
- SQL have also relationship: SQL have scan full. But GraphDB is direct relationship.

Concept && Coding

2. **Nature** :- It is distributed in Nature.



3:- Scalability :- Horizantal

My User Table growing fast User Table 1m

We can add any number of Nodes store user Data.

4: Property :- It's following Base ACID x

BASE

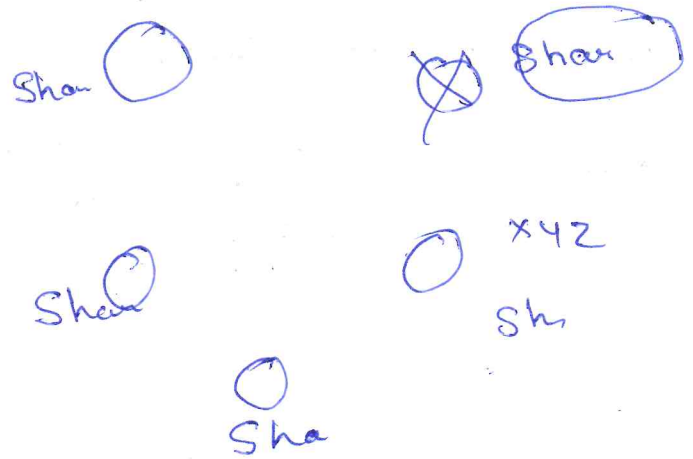
B A → Basically Avillable — highly avillable DB

S → Safe State

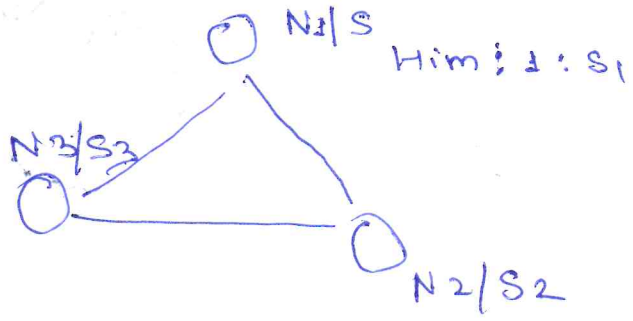
E → Eventual Consitency

Concept && Coding

BA



S - SafeState :- State of Data can be change without Interaction



E →

Query

← stale

Que after Tim →

← Lastest

When to Use :- ??

SQL	NoSQL
1 flexible Query functionlity	X (basic Query Search or In Advance, what Colum you need Search capbility.
2 Data is Relation Nature (Parent child)	
3: Dat Integrity (ACID)/Consistency	ACID X NoSQL is Developed to handle Large of Data. • High Avalibily / High Performance/Search/Query with some Consifancy.

Concept && Coding